



Kenya Climate Smart
Agriculture Project



CLIMATE SMART TECHNOLOGIES, INNOVATIONS AND MANAGEMENT PRACTICES FOR APICULTURE VALUE CHAIN

Training of Trainers' Manual



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March 2020

Disclaimer

The information presented in this manual is for advisory use only. Manual users should verify site specific appropriateness with regard to agro-climatic zones, farming system and the value chain.

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Published by

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ISBN

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Foreword

Kenya Climate-Smart Agriculture Project (KCSAP) tasked the Kenya Agricultural & Livestock Research Organization (KALRO) with the implementation of the project Component 2, on ‘Strengthening Climate-Smart Agricultural Research and Seed Systems’. The component activities are geared towards the development, validation, adoption and delivery of context specific climate smart agriculture (CSA) technologies, innovation and management practices (TIMPS). The other responsibility was development of sustainable seed production and distribution systems for priority value chains to enhance availability and access to seed, breeds and fingerlings by target beneficiaries under Components 1 (Up scaling Climate-Smart Agricultural Practices). Against this background, KALRO and her NARS partners have developed, validated and availed CSA TIMPS for dissemination and adoption. The TIMPS have further been unpacked during the development of Training of Trainers (ToT) Manuals for use in training public and private extension service providers and lead farmers.

The ToT Manuals are instructional guides to be used for teaching and learning step-by-step procedures of implementing CSA innovations for each of the 13 value chains being addressed. The training content is drawn from the CSA TIMPS that support respective value chains. The content are arranged in progressive modules supported by extensive information from research information and background data drawn from the TIMPS. Their relevance are based on the needs teased out of the value chains and the project objectives. The ToT Manuals training design takes into consideration the delivery system, the partners and their roles, the duration of training and logical flow of the sessions. Similar content requiring similar delivery systems are grouped together while the roles of the partners are tapped in the training and planning of the training sessions.

The Manual is divided into modules, which have a uniform outline that ensures every aspect of the TIMPs are fully covered in way that the trainees can absorb and relate to. Various delivery methods are deployed and where possible demonstrations and practical work are incorporated to enable the trainees learn by participating in the actual field activities. Furthermore, to ensure that the training across various groups is standardized, trainers guidelines, detailed descriptions of the trainees, program, training methods and a training evaluation have been provided in the manual. Adhering to these guidelines, therefore, enables possibility to replicate the training in several locations without loss of details regardless of whether conducted by different trainers.

It is highly advised that the ToT Manuals should be used in conjunction with the respective value chains’ TIMPs documents and facts sheets in order to provide valuable resource for both public and private extension service providers. The use of this Manual is expected to enable achievement of the envisaged ‘Triple Wins’ of increased productivity, enhanced resilience and reduction of greenhouse gases emission.

I am greatly indebted to the value chain leaders and all those who participated in the preparation of the Manual, which is expected to herald a new way of delivering training content in a changing agricultural environment.

Eliud K Kireger, PhD, OGW

Director General, KALRO

Preface

The Kenya Climate-Smart Agriculture Project (KCSAP) is a Government of Kenya project with support from both the World Bank and the government. It is a five-year project implemented in 24 counties, mainly in the arid and semi-arid lands (ASALs), at a cost of Ksh. 25B. The project development objective (PDO) is “to increase agricultural productivity and build resilience to climate change risks in the targeted smallholder farming and pastoral communities, and in the event of an Eligible Crisis or Emergency, to provide immediate and effective response.” This objective is to be achieved through the implementation of five key components, which are 1) Up scaling Climate-Smart Agricultural Practices, 2) Strengthening Climate-Smart Agricultural Research and Seed Systems, 3) Supporting Agro-weather, Market, Climate, and Advisory Services, 4) Project Coordination and Management and 5) Contingency Emergency Response.

Component 1 involves facilitating the empowering of farmers and communities to adopt technologies, innovations and management practices (TIMPs) to achieve the Climate Smart Agriculture (CSA) triple-wins of; increased productivity, enhanced resilience (adaptation), and reduced Greenhouse gas (GHG) emissions (mitigation). Component 2 is charged with the responsibility of providing the TIMPs. Therefore, it supports the development, validation, and adoption of context specific CSA TIMPS to target beneficiaries under Components 1 and 3 as well as development of sustainable seed production and distribution systems.

To catalyze uptake of TIMPs, Kenya Agricultural & Livestock Research Organization (KALRO) in conjunction with partners in the National Agricultural Research Systems (NARS) and Consultative Group for International Agricultural Research (CGIAR) compiled inventories of TIMPs for each of the 13 prioritized value chains (cassava, green grams, sorghum, millet, pigeon peas, bananas, tomatoes, potatoes, apiculture, indigenous chicken (meat and eggs), dairy (cattle and camel), red meat (cattle, sheep and goats) and aquaculture and 3 cross cutting value chains (natural resource management, pastures and fodder and animal health). The TIMPs were categorized into those ready for upscaling, those that needed validation and gaps that required further research. Training of Trainers’ (ToT) manuals focusing on TIMPs that are ready upscaling for each of the value chains were subsequently developed and form the basis of training county extension staff, service providers and lead farmers. They are in turn expected to cascade this training to beneficiaries in the targeted smallholder farming, agro-pastoral and pastoral communities in the 24 project counties of Marsabit, Isiolo, Tana River, Garissa, Wajir, Mandera, West Pokot, Baringo, Laikipia, Machakos, Nyeri, Tharaka Nithi, Lamu, Taita Taveta, Kajiado, Busia, Siaya, Nyandarua, Bomet, Kericho, Kakamega, Uasin Gishu, Elgeyo Marakwet and Kisumu.

KALRO having the mandate of implementing of activities under Component 2, has been instrumental in using its information resources and those of partners and collaborators to come up with the inventories of TIMPs and corresponding ToT Manuals. The use of

these information resources coupled with the accompanying training and the contribution of the other project components, will go a long way in enabling the KCSAP to meet its development objective.

The National Project Coordination Unit is grateful to all who participated in the development and production of this *Training of Trainers' (ToTs) Manual for Apiculture Value Chain*. It is my hope that counties and other users will put this resource to good use as they transform and reorient their agricultural systems to make them more productive and resilient while minimizing GHG emissions under the new realities of a changing climate.

Francis Muthami

National Project Coordinator

Kenya Climate-Smart Agriculture Project

LIST OF ABBREVIATIONS

AEZ	Agroecological zone
ASALs	Arid and Semi-Arid Lands
CTT	Core Team of Trainers
GAP	Good Agricultural Practices
LFs	Lead Farmers
MT	Metric tons
ToT	Training of Trainers
VMGs	Vulnerable and Marginalized Groups



PART 1

This part consists of four sections which include:

Section 1: Background

Section 2: Module Training Content

Section 3: Training Duration

Section 4: Facilitator Guidelines.

SECTION 1: BACKGROUND

1.1 The Role of Aquaculture in the Kenyan Economy

The livelihoods of most people in rural Arid and Semi Arid Lands (ASALs) of Kenya which is as over 80% of Kenya's land mass is endowed with favourable environment and rich in biodiversity and is ideal for apiculture. This is also an area which supports a population of over 30%, which is mainly engaged in small-scale farming and is often vulnerable to food insecurity. The vulnerable and marginalized groups (VMGs) are greatly affected as they have no access to economic resources. Beekeeping has been identified as a viable agricultural enterprise that could alleviate poverty and sustain rural employment in the ASALs. Beekeeping is widely considered as one of the poverty-alleviation strategies both by the Kenyan Government and other players supporting rural development in Kenya. Unlike other agricultural practices, beekeeping can be undertaken with minimal infrastructure, little capital and easy-to-learn skills hence providing an excellent opportunity for diversifying agricultural production.

1.2 The Role of Apiculture in Food and Nutrition Security

It is estimated that Kenya produces between 15,000 and 25,000 MT of honey as well as 1000 and 5,000 MT of beeswax annually. This is a relatively low amount considering the estimated potential of 100,000 and 10,000 MT, honey and beeswax respectively.

Beekeepers are dependent on the use of low productive traditional hives which results in poor quality honey. This scenario has forced beekeepers to sell their honey through an exploitive chain of agents and consequently low prices while consumer prices are higher. Thus, the challenge for policy makers and development partners aiming to support pro-poor economic development is how to stimulate the honey value chain for sustainable incomes and improved livelihoods. Technologies, innovations and management practices (TIMPs) have been developed through research to address some of the challenges along the apiculture value chain. Adoption of the TIMPs is likely to increase honey production and consumption for improved nutrition and health.

1.3 Apiculture as a Climate Smart Innovation

Technologies, innovations and management practices (TIMPs) have been developed through research to address some of the challenges in the apiculture value chain. The TIMPs focus on increased productivity, enhanced resilience to climate change and reduced greenhouse gas (GHG) emissions. Adoption of the TIMPs will increase honey production in the ASALs and other areas where climate change constraints crop and livestock productivity.

1.4 Objective of the training

The purpose of this training is to provide beekeepers trainers with knowledge and skills on beekeeping technologies and management practices. These include beekeeping

and beehive technologies, bee foraging and ecosystem conservation, housing of bee colonies and types of housing technologies used, harvesting, post harvesting handling and processing, integrated bee pastures, strategic use of honeybees for crop pollination, social issues (gender) and cost benefit analysis.

SECTION 2: TRAINING CONTENT

2.1 Orientation of the Modules

The training content is organized into 8 modules spread over a period of 3 hours 55 minutes. The modules are orientated to ensure adoption and upscaling of apiculture TIMPs to improve productivity, resilience and mitigation of harmful greenhouse gases. The purpose of these modules is to enhance the knowledge and capacities of trainers in understanding and applying climate smart agricultural practices in apiculture.

2.2 Module outline

Each of the 8 modules has a similar outline consisting of 8 parts. These are:

1. Introduction to the module – context and background to training needs, knowledge and skills gaps being addressed
2. Module learning outcomes – what trainees are expected to learn
3. Module target group-trainee categories
4. Module users –facilitators
5. Module duration –minimum number of hours of exposure to materials
6. Module summary –sequence of sessions, training methods, materials and duration
7. Facilitator’s guideline –detailed sessions, training methods, materials and session guides
8. Participant’s handouts – detailed notes and reference materials for trainees.

The outline for each of the 8 modules in Table 1.

Table 1: Outline of the eight modules

No	Module Name	Need Addressed	Expected Training Outcomes	Duration
1.1	Beekeeping and Beehive technologies	Lack of information on choice of hives in different ecological zones	Well sited apiaries and appropriate choice of hive Enhanced production	4 hours 40 minutes
1.2	Bee foraging and ecosystem conservation	Dwindling bee forage plants specific to different AEZs,	Major bee forage plants identified and multiplied per county. Better community/ government forest management	2 hour 40 minutes
1.3	Housing of bee colonies and types of housing technologies used	Absconding of bee colonies due to temperature variation, and effects of climate change	Beekeepers to be able to choose hives types and housing systems suitable for their needs. Proper siting of apiaries and improved bee pests and disease management	3 hours 40 hours
1.4	Harvesting, post harvesting handling and processing	Poor quality, honey wastage and poor packaging materials	High quality honey and proper packaging	4 hours 35 minutes
1.5	Integrated Bee Pastures	Increased habitat degradation in most beekeeping areas	Reclamation and increase in forest cover	3 hours 45 minutes

1.6	Strategic use of honeybees for crop pollination	Increased crop yields and quality	Strategic use of honeybees for crop pollination in farmers crop fields	4 hours 10 minutes
1.7	Social issues	Low involvement of women, youth and the vulnerable in honey value chain	More involvement of women, youth and the vulnerable in honey value chain	4 hours 15 minutes
1.8	Beekeeping Cost benefit analysis	Low incomes and unprofitable enterprises	High Incomes	3 hours 10 minutes
Total Duration				30 hours 55 minutes

SECTION 3: TRAINING DESIGN

3.1 Delivery System

The delivery system designed for this training consists of two stages:

a) Establishment of a team of facilitators

- ⇒ A Core Team of Trainers (CTT) trains farmer trainers (service providers, public and private extension agents and lead farmers) as facilitators of a Trainer of Trainers (ToT) course. This is done using this manual and modules contained therein.
- ⇒ Each of the Mater trainers will facilitate farmers to acquire knowledge and skills in facilitating Farmer-led Field and Business Schools through practical demonstrations through PowerPoint presentations, videos, practical demonstrations and provision of illustrated training notes/brochures.
- ⇒

b) Upscaling –This will be done through farmer field and business school (FFBS) training and demonstrations, participatory on-farm validation trials and demonstrations, ASK and NARS shows and by selecting lead farmers (LF) to be trained in facilitation skills

3.2 Partners and Their Roles

The partners envisioned in this training plan are:

- 1. Core Team of Trainers** – Master trainers drawn from KALRO, National Museums of Kenya, and Universities involved in Apiculture research. This team will offer backstopping accordingly to farmers' trainers during the first year of trainings. They should also be available in the evaluation of the first round of trainings.
- 2. County Government Department of Agriculture and Livestock** –Trainers and their supervisors referred to as County Coordination Teams (CCT) will take the role of LF trainers, mentors and coordinators at sub county level. They will assist FFBS's to form partnerships and networks with stakeholders for sustainability.
- 3. Lead Farmer Networks**-Association of LFs in the counties are expected to take up farmer trainings and up scaling in the future. Lead farmer networks and groups will conduct exchange visits to learn best practices in other project implementing counties.
- 4. Private Sector Service Providers** – Inputs suppliers, financial and business development service providers, market players and processors will partner and support growth of individual and/or beekeeping groups.

3.3 Training duration

The proposed initial ToT course for Master trainers for 8 modules in the Apiculture value chain shall take a total of 30 hours 55 Minutes within a six (6) day training period. This does not include break hours of mid-morning, afternoon and lunch breaks.

3.4 Logic of Design and Flow of Sessions

The logic of design and flow of each module is that the facilitator while paying attention to the proposed methods and sessions guidelines shall: (1) Introduce the module; (2) Draw out the participant's expectations; (3) Relate participants' expectations with r learning outcomes; (4) Explore the concept and content, switching to different methods of delivery of the content (e.g. group exercise, brainstorming, excursions, plenary discussions, role plays among others) as the session progresses; (5) Review the module at the end using participatory approaches like on participant reads one summary message and its application; and, (6) Distribute the relevant participant's hand outs.

SECTION 4: FACILITATOR GUIDELINES

4.1 Preparation of Training Materials

The training materials suggested require adequate preparations and should be available before the actual training dates. Furthermore:

- a) The facilitators should familiarize themselves and internalize the guidelines provided by this manual prior to the training;
- b) The stationery required should be available within the training institution 3 days before the training. These include name tags, writing materials, paper punch and medium size box files for participants' handouts filling;
- c) Flip charts and good quality felt pens could be used interchangeably with LCD projections. Each participant will require one felt pen while the trainers will require two sets of felt pens;
- d) Visual aids like field equipment and tools should also be arranged in time before the sessions start;
- e) There should be adequate copies of participants' hand outs (one per participant) to be distributed at the end of each session or as may be suitable; and
- f) Copies of the modules should be distributed at the end of each module.

4.2 Preparation of Training Venue and Sites

The training venue will include the training room, field demonstration sites and market areas.

- a) **Training Room** – should have adequate space for 25 participants seated in a semi-circle or U shape arrangement ensuring access and unobstructed view of the front. There should be adequate space for a desk and seats for 3 trainers preferably at the sides or at the back of the training room. There should also be a desk for the trainer, their training materials and LCD projector, a flip charts holder and white wall to act as a projector screen.
- b) **Demonstration Site** – Should be within walking distance within an apiary site.
- c) **Market Sites** – these include honey retail outlets (Kiosks, stalls, shops and supermarkets), wholesale and aggregation points and processing sites if available. The operators should be informed in advance about the visits. These should not be very far away preferably less than 10 minute drive distance

4.3 The Trainees

The trainees who will participate are extension officers, lead farmers, educators/universities, and researchers who are subject matter specialist with an elaborate training background in extension and advisory services. They will be drawn from public and private sector based on considerable experience in training farmers but with minimal facilitative advisory or technology transfer approaches. The trainer should, therefore, act more of a facilitator than a lecturer and draw out and build on their knowledge, skills and experience that they shall bring in. The facilitator's role will be to facilitate and listen and let them feel like equals to each other and the CTT team members.

4.4 Training Program

The training program consists of the actual training modules. Health breaks should be considered when drawing the training program. The training program should preferably be based on the outline presented in **Annex 1** to allow flow of ideas and topics. However, should the situation demand, the sequence and day of coverage for whole or parts of the modules can be modified to suit emerging requirements. The training program assumes that the trainees report on Sunday evening as the first day and leave ten days later on Friday afternoon or Saturday morning.

4.5 Training Methods

The training methods proposed for each session are suitable for adult learners and appropriate for addressing knowledge, skills and attitudes of the participants. The choice of the methods has been informed by the competency issues being addressed, time available and experiences of the authors of this manual. Depending on time available, the facilitator can modify these training methods but as a golden rule, no presentation by the facilitator should take more than 30 minutes continuously. Presentations should be separated by the other participatory training methods.

The list of available training methods and activities is presented in Table 2.

Table 2: Training methods

Training Method	Description of Method
Plenary presentations	Use of Power Point or flip charts and plenary discussions in situations where knowledge and opinion or consensus is required
Group exercises, buzz groups, visits and brainstorming sessions	To be considered where skills are an issue requiring sharing and trying

Role plays and problem-solving exercises	Plenary discussions have been considered as training methods where attitude is an issue
On-farm practical demonstration	To be considered where hand-on practical skills are acquired through sharing and demonstration

4.6 Planning Schedule and Guidance for ToT preparation

While planning for this training, the CTT leader should ensure the following before the training:

Table 3: Training Preparation

Duration to Training	Activities to be Done
Six weeks	Recruit master trainers, compose CTT, identify the practical demonstration sites
Four weeks	Send out invitation letters to participants and special guests detailing purpose, venue and program. Follow up on demonstration sites. Brief CTT members
Two weeks	Confirm names of participants; reproduce training materials for facilitators and package, confirm preparedness of the field sites to be visited. Hold briefing of CTT members to finalize training plan. Confirm availability of special guests if any
Four days	Confirm training sites preparedness, prepare sitting arrangements, and brief assistants
One day	Arrange training room furniture, place materials, equipment and stationery on the tables. Arrange for reception of trainees at residence proposed
On first day	Arrange for the reception of trainees at the training venue. Ensure climate setting is done before the course is officially opened. This includes: (i) registration, (ii) welcome to the venue by the host; (iii) elaborate introduction of CTT and participants; setting of ground rules; and (iv) formation of groups

4.7 Evaluation of Training

About an hour has been allocated for planning for way forward and evaluation of the ToT on the last day of the training. This is as presented in the program just presented in section 4.4.

The evaluation strategy should take two directions. The first being **the individual evaluation** where each of the trainees evaluates the training using provided evaluation forms without discussing with any other trainee (Table 4). The individual evaluation forms are then collected and analysed by the CTT members.

Table 4: Individual Sample Evaluation Form

Aspect / Module	Rating		
	Very useful (3 marks)	Useful (2 marks)	Least useful (1 mark)
Beekeeping and Beehive technologies			
Bee foraging and ecosystem conservation			
Housing of bee colonies and types of housing technologies used			
Harvesting, post harvesting handling and processing			
Integrated Bee Pastures			
Strategic use of honeybees for crop pollination			
Social issues			
Beekeeping Cost benefit analysis			

The second evaluation is the trainee's group evaluation. They retreat to one room and elect a chair and a secretary. They are then asked to objectively and constructively evaluate the training in about 45 minutes in the absence of the CTT members. They then present their evaluation to the CTT members and as they present, the CTT members should only give points of clarifications if any misunderstanding occurred but should not try to be defensive. The CTT members then use the two evaluation results to write a report highlighting aspects that went on well and can be replicated, challenges that were encountered, and opportunities for future improvement of the ToT.

4.8 Facilitator's Training Notes and Reference Materials

4.8.1 Key references

Two key references should be provided for each module plus a list of other relevant publications for reference. The detailed list of all references is summarized in Annex 2

4.8.2 Guide on the use of the information

The trainers will be advised to issue farmers with at most 2 publications for each of the training sessions. This is because if they go away with 10 publications in one visit, they may be overwhelmed with the material load and thus limit knowledge uptake. Also, some will just take away as many as they can if allowed.

The list of all individual publications will be stored and availed as electronic copies – mainly PDFs. The service providers are strongly advised to keep these electronic copies on a memory stick, CD or portable hard drive to enable farmers easily access them and if necessary, print any of them out at a local internet café.

Trainers will be advised to issue one General Apiculture Farming Manual to be accompanied by two other publications e.g. information sheets, brochures, factsheets and poster. With subsequent training modules, they can develop their collection of publications.



PART II

This part presents the content of 8 modules namely: (1) Beehive technologies, (2) Bee foraging and ecosystem Conservation; (3) Housing of bee colonies and types of housing; (4) Harvesting, post harvesting handling and processing; (5) Integrated bee pastures; (6) Strategic use of honeybees for crop pollination; (7) Cost benefit analysis; and (8) Social issues (gender).

The 8 modules are divided into the following sections:

- 1.1. Introduction to the Module
- 1.2. Module Learning Outcomes
- 1.3. Module Target Group
- 1.4. Module Users
- 1.5. Module Duration
- 1.6. Module Summary
- 1.7. Facilitator's Guidelines
- 1.8. Participants' Handouts and Training References



MODULE 1

UP-SCALING HIVE TECHNOLOGIES

1.1 Introduction to the Module

This module aims at addressing issues of beehive technologies and up scaling to reach the larger community for increased productivity, promote technologies and innovations and enhance livelihoods of the beekeeping community. This module concentrates on the hive features that affect honey productivity.

1.2 Module learning outcomes

By the end of the module training the following training outcomes must be achieved:

- Importance of beekeeping explained
- Species and races of bees identified
- Common types of beehive in Kenya identified and discussed
- Advantages and disadvantages of different beehives explained
- Production levels of different hives appreciated.

1.3 Module Target groups

This module targets service providers who include county extension staff and private service providers and lead farmers

1.4 Module Users

This module is intended for use by trainers who are members of the Core Team of Trainers (CTT) and Farmer Trainers. The trainer should thoroughly familiarize themselves with the participant's handouts and training reference materials.

1.5 Module Duration

The Module is estimated to take 4 hours 40 minutes.

1.6 Module summary

Up scaling of beehive Technologies			
Sessions	Training Methods	Training Materials	Time
1.6.1 Introduction and leveling expectations	• Presentation • Discussion	• Felt pens, masking tapes and glue • Flip charts • LCD projector • Handout	20 minutes
1.6.2 Importance of beekeeping	• Group exercise • Presentation of group exercise • Discussions	• Flip charts • Felt pens • PowerPoint Projector • Handout	50 minutes
1.6.3 Species and races of bees	• Group exercise • Presentation of group exercise • Discussions	• Flip charts • Felt pens • Participants' handouts	1 hour
1.6.4 Common types of beehives in Kenya	• Presentation	• Flip charts • Felt pens • Handout	1 hours
1.6.5 Advantages and disadvantages of different beehives	• Discussion • Plenary presentation	• Flip charts • Felt pens • PowerPoint projector • Handout	30 minutes
1.6.6 Hive Production levels	• Group exercise • Plenary presentation	• Flip charts • Felt pens • PowerPoint projector • Handout	40 minutes
1.6.7 Module Review	• Power-Point presentations • Plenary discussion	• Flips charts • PowerPoint projector • Handout	20 minutes
TOTAL			4 hours 40 minutes

1.7 Facilitator's Guidelines

Up scaling of beehive technologies	
1.7.1 Introduction and Levelling Expectations (20 minutes)	Session Guide
Introduction <i>(The facilitator introduces the module and invites trainees to introduce themselves and state their expectations).</i> <i>The facilitator then presents module objectives and levels out expectations</i> Module Objectives • By the end of the module trainees should be able to : • Explain what beekeeping is. • Identify honeybee species and races in Kenya • Identify and discuss types of hives used in Kenya • Explain the advantages and disadvantages of different beehives • Appreciate the different production capacities of different hives	• Summarize trainees expectations using cards or any appropriate method. • PowerPoint presentation • Distribute Participants' Handout on Module Objectives.
1.1.2 Importance of beekeeping (50 minutes)	
Plenary Presentation (30 minutes). <i>(The facilitator makes a presentation on the importance of beekeeping)</i> • Social and economic importance of bees Plenary discussion (30 minutes) Let the participants recall what they learned and discuss any issues that may arise	PowerPoint presentation and discussion Distribute Participants' Handout

Honeybee species and races of bees (30 minutes)	
Plenary Presentation (30 minutes) <i>(The facilitator should be able to guide the trainees in identifying different bee species i the region.)</i> • Honeybee species • Races of bees • Distinct features of different bee races • Productivity of each bee race • Distribution of bee races in Kenya • Ecological and conservation of bee races Plenary discussion (30 minutes) Let the trainees recall what they learned and discuss any issue that may arise	PowerPoint presentation and discussion Distribute Participant's handout
1.7.4 Common types of beehives in Kenya (60 minutes)	
Plenary Presentation (30 minutes) <i>(The facilitator guides the trainees in distinguishing different bee hives).</i> • Different types of common beehives in Kenya • Comparisons of different hive types • Innovation/Improved beehives under climate change and variability The image shows a wooden box hive on the left and a cross-section diagram of a beehive on the right. The diagram labels various parts: 'Entrance hole', 'Queen bee', 'Worker bees', 'Honey comb', 'Bee bread', and 'Bee larvae'. It also includes dimensions: '1.5m' for the length, '0.5m' for the width, and '0.3m' for the height. Discussion (30 Minutes) After the presentations allow trainees to raise any issues and discuss them	PowerPoint presentation and discussion Distribute Participants' handout

1.7.5 Advantages and disadvantages of different beehives (20 minutes)	
Plenary Presentation (20 minutes) <i>(The facilitator should be able to lead the trainees in highlighting the advantages and disadvantages of each type of beehive)</i> • Advantages and disadvantages of different beehives • Gender friendliness of different hive types Discussion (10 minutes) After the presentations allow trainees to raise any issues and discuss them	PowerPoint presentation and discussion Distribute Participant's handout
1.7.6 Hive Production levels (50 minutes)	
Plenary Presentation (30 minutes) <i>(The facilitator leads the trainees in aspects honey production and production levels of different hive types)</i> • Honey production seasonal calendar • Productivity of different hive types • Volumes of honey produced per region Discussion (20 minutes) After the presentation, allow trainees to raise any issues and discuss them.	PowerPoint presentation and discussion Distribute Participants' handout
1.7.7 Module review (20 minutes)	
Plenary Presentation (20 minutes) <i>(The facilitator leads the trainees in presenting their views on each of the sessions covered under this module)</i> • Review together with the trainees the main points Beekeeping and Beehive technologies • What new things did you learn from this module? • What are some of the issues that you have become more aware of? • What questions do you still have?	Recap of the key take home points using any of the following participatory methods: • Q & A session • Discussions • Any other Distribute participants' handout on module review

1.8 Training Notes and Reference Materials

1.8.1 Participant's Handout

1. Handout on up scaling of beehive Technologies

1.8.2 Training Reference Materials

1. Beekeeping in Kenya – Beginners guide (Ministry of Livestock Development 2011)
2. *Beekeeping: a beginner's guide* 1997. Baraka Agricultural College, Baraka, Kenya. (Carroll, T.)
3. *Bees for wealth and health: (Brochure)* Ministry of Agriculture & Rural Development, 2000 Nairobi, Kenya.

MODULE 2

BEE FORAGE AND ECOSYSTEM CONSERVATION

2.1 Introduction to the Module

Beekeeping in Kenya is majorly practised in the ASALs by small scale farmers with low yields. This is attributed to several factors which include poor production methods, entrepreneur attitude, low adoption of appropriate technology and inadequate training in GAP.

2.2 Module learning outcomes

This module aims at developing bee forage nurseries and promotion of conservation of the same. By the end of the module, trainees will have acquired knowledge and skills on:

- Bee forages in different AEZs
- Bee forage conservation
- Bee pollination and its significance

2.3 Module Target Group

This module targets service providers who include county extension staff and private service providers and lead farmers.

2.4 Module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT) and farmer trainers. The trainees using this module should thoroughly familiarize themselves with the participants' handouts and training reference materials.

2.5 Module duration

The Module is estimated to take 2 hours 40 minutes

2.6 Module summary

Bee forage and ecosystem conservation			
Sessions	Training Methods	Training Materials	Time
2.6.1 Introduction to the module and leveling the expectations	• Presentation • Discussion	• Felt pens, masking tape or sticker glue, notebooks and pens • PowerPoint Projector Slides • Handout	20 minutes
2.6.2 Types of bee forages in different AEZs	• Plenary presentation • Group exercise Summary presentation	• PowerPoint Projector • Exercise handouts • Flip charts • Felt pens, • Participants' handouts	30 minutes
2.6.3 Bee forage conservation	• Plenary presentation • Group exercise Summary presentation	• PowerPoint Projector • Exercise handouts • Flip charts • Felt pens • Participants' handouts	30 minutes
2.6.4 Bee pollination and its significance	• Group Exercise • Plenary discussions	• Flip charts • Felt pens, • Participants' handouts	1 hour
2.6.5 Module review	• Participants' questions and comments • Facilitator's summary	• Flip charts • Felt pen • Participants' Handout on module review	20 minutes
TOTAL			2 hour 40 minutes

2.7 Facilitator's Guidelines

Bee Forage and Ecosystem Conservation	
2.7.1 Introduction and Levelling Expectations (20 minutes)	Session Guide
Introduction <i>(The facilitator welcomes trainees to the module Bee Forage and Ecosystem Conservation, then introduces themselves and state their expectation for the module).</i> <i>The facilitator then presents the module objectives.</i> Module Objectives By the end of the module trainees should be able to: • Describe and explain the different types of bee forage plants • Understand and explain the value of trees and their conservation in the environment • Understand and explain bee forage preferences	Summarize Participants' "Expectations" and display them PowerPoint presentation Distribute Participants' Handouts on module objectives.
1.1.2 Types of bee forages in different AEZs (30 minutes)	
Plenary Presentation (20 minutes) <i>(The facilitator should describe the different vegetation cover and tree abundance in different AEZs)</i> The facilitator makes a PowerPoint presentation on types of bee forages in different AEZs The presentation should highlight: • The AEZs, • preferred bee forage plants in the environment, • the challenges faced under the current climate change and weather variability Discussion (10 minutes) Let the trainees recall what they learned and discuss any issue that may arise	PowerPoint presentation and discussion Distribute Participant's handout

1.1.3 Bee forage conservation (30 minutes)	
Plenary Presentation (20 minutes) <i>(The facilitator guides the trainees in identifying ways of conserving bee forage to enhance honey productions).</i> • Types of tree plants source of nectar and pollen • Seasonal variations • Threat of climate change and weather variability • Floral calendar Discussion (10 minutes) Let the trainees recall what they learned and discuss any issues that may arise	PowerPoint presentation and discussion Distribute Participants' handout
2.7.4 Bee pollination and its significance (30 minutes)	
Plenary Presentation (20 minutes) <i>(The facilitator leads the trainees in understanding Bee pollination aspects and its importance).</i> The presentation should highlight: • Bees as pollinators • Crops/food pollinated by honeybee • Quantity of food produced as a result of bee pollination Discussion (10 Minutes) After the presentations allow trainees to raise any issues and discuss them	PowerPoint presentation and discussion Distribute Participant's handout
2.7.5. Module review (20 minutes)	

Plenary Presentation (20 minutes) <i>(The facilitator leads trainees present their views on each of the sessions covered under this module).</i> • Review together with the trainees the main points about Bee Forage and Ecosystem Conservation • What new things did you learn from this Module? • What are some of the issues that you have become more aware of? • What questions do you still have?	Recap of the key take home points using any of the following participatory methods: • Q & A session • Discussions • Any other Distribute Participants' handout on module review
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2.8 Training Notes and Reference Materials

2.8.1 Participants' Handouts

1. Handout on bee forage and ecosystem conservation

2.8.2. Training Reference Materials

1. Beekeeping in Kenya – Beginners guide ((Ministry of Livestock Development 2011)
2. *Beekeeping: a beginner's guide* 1997. Baraka Agricultural College, Baraka, Kenya. (Carroll, T.)
3. *Bees for wealth and health: (Brochure)* Ministry of Agriculture & Rural Development, 2000 Nairobi, Kenya.

MODULE 3

HOUSING OF BEE COLONIES AND TYPES OF HOUSING TECHNOLOGIES

3.1 Introduction to the Module

This module is designed for use in training facilitators on housing of bee colonies and types of housing. There is a discontent between available bee colonies, honey production and bee forage availability.

3.2 Module learning outcomes

This module aims at training the facilitators to empower farmers to increase productivity and improve marketing for enhanced incomes. This will be done through enhanced market assessments that will enable the farmers engage in sustainable beekeeping.

By the end of the module training the following training outcomes should be achieved:

- Need for housing of apiaries in Kenya understood
- Siting of apiaries appreciated
- Importance of regular beehive inspection and hygiene realized
- Need for disease and pest management surveillance appreciated
- Need for supplementary feeding recognized

3.3 Module target group

This module targets service providers who include county extension staff and private service providers and lead farmers

3.4 Module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT) and farmer trainers. The trainees using this module should thoroughly familiarize themselves with the participants' handouts and training reference materials.

3.5 Module duration

The Module is estimated to take 3 hours 40 minutes

3.6 Module summary

Housing of bee colonies and types of housing technologies			
Sessions	Training Methods	Training Materials	Time
3.6.1 Introduction and leveling expectations	• Presentation • Discussion	• PowerPoint Projector • Participants' Handouts • Felt pens, masking tapes and glue • Handout	20 minutes
3.6.2 Importance of housing of apiaries	• Plenary presentation and discussions	• Flip charts • Felt pens • PowerPoint Projector • Handout	30 minutes
3.6.3 Siting of apiaries	• Buzz activity • Presentation discussion	• PowerPoint Projector • Flip charts • Felt pens • Handout	30 minutes
3.6.4 Regular beehive inspection and hygiene	• Presentation • Plenary discussion	• PowerPoint Projector • Flip charts • Handouts • Notebooks	40 minutes
3.6.5 Disease and pest management	• Market walk • Plenary discussions of data collected	• Samples for assessment • Handout • Checklist/ tools • Flip charts	1 hour
3.6.6 Supplementary feeding	• Plenary discussion • Plenary presentation	• PowerPoint Projector • Flip charts • Analysis template • Participants' handouts	20 minutes

3.6.7 Module review	• Participants' Facilitator's summary	• Flip charts • Felt pens • Participants' Handout for module review	20 minutes
TOTAL			3 hours 40 minutes

3.7 Facilitator's Guidelines

Housing of Bee Colonies and Types of Housing Technologies	
3.7.1 Introduction and Levelling Expectations (20 minutes)	Session Guide
Introduction <i>(The facilitator welcomes trainees to the module and invites them to introduce themselves and state their expectation for the module).</i> The facilitator then presents the module objectives. Module Objectives By the end of the module the trainee should be able to: • Understand of housing of apiaries in Kenya. • Appreciate Siting of apiaries. • Realize importance of regular beehive inspection and hygiene. • Disease and pest management. • Recognize the need for supplementary feeding.	Summarize Participants' "Expectations" and display them PowerPoint presentation Distribute Participant Handouts on Module Objectives.
3.7.2 Importance of housing of apiaries (30 minutes)	

3.7.5 Disease and pest management (60 minutes)	
Plenary Presentation (40 minutes) <i>(The facilitator leads the trainees in highlighting the importance of control of diseases and pests in bee colonies)</i> • Different types of pests and diseases affecting bee colony • Control measures • Hive hygiene builds a strong colony Discussion (20 minutes) After the presentations allow trainees to raise any issues and discuss them	PowerPoint presentation and discussion Distribute Participants' handout
3.7.6 Supplementary Feeding (20 minutes)	
Plenary Presentation (10 minutes) <i>(The facilitator leads the trainees in aspects supplementary feeding during drought period).</i> The presentation should highlight: • Seasonal variations and associated challenges • Composition of supplementary feed • Colony sustenance – Discussion (10 minutes) After the presentations allow participants to raise any issues and discuss them.	PowerPoint presentation and discussion Distribute Participants' handout
3.7.7 Module review (20 minutes)	

<i>(The facilitator leads trainees in presenting their views on each of the sessions covered under this module).</i> • Review together with the trainees the main points in Housing of Bee Colonies and types of Housing Technologies • What new things did you learn from this Module? • What are some of the issues that you have become more aware of? • What questions do you still have?	Recap of the key take home points using any of the following participatory methods: • Q & A session • Discussions • Any other Distribute handout on module review Distribute handout on module review
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3.8 Training Notes and Reference Materials

3.8.1 Participant's Handouts

1. Handout on Housing of Bee Colonies and Types of Housing Technologies

3.8.2. Training Reference Materials

1. Beekeeping in Kenya – Beginners guide ((Ministry of Livestock Development 2011)
2. *Beekeeping: a beginner's guide* 1997. Baraka Agricultural College, Baraka, Kenya. (Carroll, T.)
3. *Bees for wealth and health: (Brochure)* Ministry of Agriculture & Rural Development, 2000 Nairobi, Kenya.

1.1.1

MODULE 4

HARVESTING, POST HARVESTING HANDLING AND PROCESSING

4.1 Introduction to the module

This module is designed for training facilitators in honey harvesting, post harvesting handling and processing in a changing variable climate. Farmers use unconventional methods for harvesting and processing honey. Good honey quality is critical to meet market requirements.

4.2 Module learning outcomes

By the end of the module, participants will have acquired knowledge and skills on:

- Methods of honey harvesting described and explained
- Honey processing understood and explained
- Post-harvest handling understood and explained.
- Packaging and labelling understood and explained.

4.3 Module target group

This module targets service providers who include county extension staff and private service providers and lead farmers.

4.4 Module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT) and farmer trainers. The trainers using this module should thoroughly familiarize themselves with the participant's handouts and training reference materials.

4.5 Module duration

The Module is estimated to take 4 hours 35 minutes.

4.6 Module summary

Harvesting, post harvesting handling and processing			
Sessions	Training Methods	Training Materials	Time
4.6.1 Introduction and levelling of expectations	• Buzz • Presentation	• Copy of module Objectives	20 minutes
4.6.2 Honey harvesting and equipment	• Plenary presentation • Discussions	• PowerPoint Projector, Photos • Flip charts • Participants' Handouts	20 minutes
4.6.3 Harvesting procedure	• Presentation • Field demonstration	• Soil sampling tools, Sample packages and Data sheets	55 minutes
4.6.4 Hive products	• Presentation • Discussions	• PowerPoint Projector • Participants' handouts	50 minutes
4.6.5 Honey processing and packaging	• Presentation • Practical session	• PowerPoint Projector, Colored photos • Handouts	50 minutes
4.6.6 Hive products value addition	• Group discussions and presentation	• PowerPoint Projector	50 minutes
4.6.7 Module review	• Participants' Facilitator's summary	• Participants' • Handout for module review	20 minutes
TOTAL			4 hours 35 minutes

4.7 Facilitator Guidelines

Harvesting, Processing and Post Harvesting Handling	
4.7.1 Introduction and Levelling Expectations (20 minutes)	Session Guide
Introduction <i>(The facilitator welcomes trainees to the module invites them to introduce themselves and state their expectation for the module).</i> The facilitator presents modules objectives Module Objectives By the end of the module trainee should be able to: • Describe and explain methods of honey harvesting. • Understand and explain honey processing. • Understand and explain post-harvest handling. • Understand and explain packaging and labelling.	Summarize trainees’ “expectations” and display them PowerPoint presentation Distribute Participants’ Handouts on module objectives.
4.7.2 Honey harvesting and equipment (30 minutes)	
Plenary Presentation (20 minutes) <i>(The facilitator describes the recommended methods of honey harvesting and required equipment)</i> • Honey harvesting equipment and kit, accessories, • Recommended harvesting, storage and processing Discussion (10 minutes) Let the participants recall what they learned and discuss any issues that may arise 	PowerPoint presentation and discussion Distribute Participants’ handout

4.7.3 Harvesting procedure (55 minutes)

Plenary Presentation (35 minutes)

(The facilitator guides the trainees in understanding the highly recommended method of harvesting honey).

- Harvesting kit
- Harvesting tools and accessories
- Post-harvest honey handling
- Physico chemical properties of honey



Discussion (20 Minutes)

After the presentations allow trainees to raise any issues and discuss them

PowerPoint presentation and discussion

Distribute Participants' handout

4.7.4 Hive products (50 minutes)

Plenary Presentation (30 minutes)

(The facilitator should be able to lead the trainees in understanding the different types of hive products).

- Products derived from bees
- Products derived from the beehive
- Value addition
- Market niche for the hive products

Discussion (20 Minutes)

After the presentations allow trainees to raise any issues and discuss them accordingly

PowerPoint presentation and discussion

Distribute
Participants'
handout

<i>(The facilitator leads the trainees in presenting their views on each of the sessions covered under this module. On flip charts list and summarize the key points they should emphasize when training farmers).</i> • Review together with the trainees the main points Harvesting, Processing and Post Harvesting Handling • What new things did you learn from this Module? • What are some of the issues that you have become more aware of? • What questions do you still have?	Recap of the key take home points using any of the following participatory methods: • Q & A session • Discussions • Any other Distribute participants' handout on module review
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4.8 Training Notes and Reference Materials

4.8.1 Participants' Handouts

1. Handout on harvesting, processing and post harvesting handling

4.8.2. Training Reference Materials

1. Beekeeping in Kenya – Beginners guide ((Ministry of Livestock Development 2011)
2. *Beekeeping: a beginner's guide* 1997. Baraka Agricultural College, Baraka, Kenya. (Carroll, T.)
3. *Bees for wealth and health: (Brochure)* Ministry of Agriculture & Rural Development, 2000 Nairobi, Kenya.

MODULE 5

INTEGRATED BEE PASTURES

5.1 Introduction to the Module

Bees depend on a wide variety of plants in different ecoregions. The main floral resources include pollen, nectar and resins. Pollen is their main source of food for the nourishment and colony survival. However, the diversity of plant resources they depend on is threatened by habitat degradation and modification through human factors. The most notable is tree logging for charcoal production and conversion of natural habitats for agricultural farming and development. Hence, the need to identify natural habitats and fabricated habitats that can be managed and conserved as bee pastures within the agro ecosystems for increased hive production and bee conservation. Introduction of beekeeping and butterfly farming to local communities around Arabuko Sokoke Forest and Kaya Muhaka forest provides excellent examples of how beekeepers can conserve natural habitats for improved delivery of ecosystem services and increased honey production.

Following the increased habitat degradation in most beekeeping areas, communities can be trained on how to identify key potential bee pastures areas within their own land or communally owned land where the population of bee plants can be increased through reseedling and replanting of resourceful plants. Such plants may include herbs, shrubs and trees that are rich in floral resources. The replanting of the floral resources should be guided by established flowering phenology within the target counties. Furthermore, communities should be encouraged to make nurseries of bee plants using the locally available germplasm. Climate-smart plants species such as Acacia and Aloe species should be identified and propagated for replanting and conservation.

5.2 Module learning outcomes

By the end of the module training the following training outcomes must be achieved:

- Identify bee pastures
- Describe and explain blooming flowers
- Describe and explain a mutually beneficial bee/flower relationship
- Identify and explain floral resources and flowering phenology
- Appreciate need for conservation of natural bee pastures and restoration of degraded habitats

5.3 Module target group

This module targets service providers who include county extension staff and private service providers and lead farmers

5.4 Module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT) and farmer trainers. The trainers using this module should thoroughly familiarize themselves with the participant's handouts and training reference materials.

5.5 Module duration

The duration of the module is estimated to be 3 hours 45 minutes.

5.6 Module summary

Integrated Bee pastures			
Sessions	Training Methods	Training Materials	Time
5.6.1 Introduction and levelling of expectations	• Plenary presentation	• Copy of Module Objectives	20 minutes
5.6.2 Introduction to bee diversity: managed and solitary bees.	• Plenary presentation • Plenary demonstration	• PowerPoint Projector, Photos • Flip charts • Felt pens • Participants' Handouts	45 minutes
5.6.3 Introduction to major bee floral resources and flowering phenology.	• Presentation • Field demonstration	• Handouts • Reference books	30 minutes

5.6.4 Conservation of natural bee pastures and restoration of degraded habitats.	• Presentation • Discussions	• PowerPoint Projector • Flip charts • Felt pens • Participants' handouts	30 minutes
5. 6.5 Management of bee pastures (Induced disturbance, reseeding, planting seedlings etc.)	• Presentation	• PowerPoint Projector • Flip charts • Felt pens • Participants' handouts	40 minutes
5.6.6 Threats to bee pastures and mitigation strategies.	• Presentation • Group discussions and presentation	• PowerPoint Projector • Flip charts • Felt pens • Participants' handouts	40 minutes
5.6.7 Module review	• Participants' questions and comments • Participants' Facilitator's summary	• Flip charts • Felt pens • Participants' handout on module review	20 minutes
TOTAL			3 hours 45 minutes

5.7 Facilitator Guidelines

Integrated Bee Pastures	
5.7.1 Introduction and Levelling Expectations (20 minutes)	Session Guide

Introduction <i>(The facilitator welcomes trainees to the module to introduces themselves and state their expectation for the module. The facilitator then presents the module objectives and levels out expectations).</i> Module Objectives By the end of the module the trainee should be able to: • Identify and explain floral resources and flowering phenology • Identify bee pastures • Describe and explain blooming flowers • Describe and explain mutually beneficial bee/flower relationship • Appreciate need for conservation of natural bee pastures and restoration of degraded habitats	Summarize trainees' "expectations" and display them Distribute Participant Handouts on Module Objectives.
5.7.2 Introduction to bee diversity: managed and solitary bees. (45 minutes)	
Plenary Presentation (30 minutes) <i>(The facilitator describes the recommended methods of honey harvesting and required equipment)</i> • Bee diversity • Types of bees (managed and solitary) Discussion (15 minutes) Let the trainees recall what they learned and discuss any issues that may arise	PowerPoint presentation and discussion Distribute Participants' handout
5.7.3 Introduction to major bee floral resources and flowering phenology. (30 minutes)	

5.7.6 Threats to bee pastures and mitigation strategies. (40 minutes)	
Plenary Presentation (25minutes) <i>The facilitator leads the trainees in aspects of Bee pasture depletion and its mitigation strategies.</i> • Bee pastures in different AEZs • Threats to bee pastures • Mitigation strategies to conserve Bee pastures Discussion (15 minutes) After the presentation, allow trainees to raise any issues and discuss them.	PowerPoint presentation and discussion Distribute Participants' handout
5.7.7 Module review (20 minutes)	
<i>(The facilitator leads the trainees present their views on each of the sessions covered under this module. On flip charts, list and summarize the key points they should emphasize when training farmers)</i> • Review together with the trainees the main points on Integrated Bee Pastures • What new things did you learn from this Module? • What are some of the issues that you have become more aware of? • What questions do you still have?	Recap of the key take home points using any of the following participatory methods: • Q & A session • Discussions • Any other Distribute handout on module review

5.8 Training Notes and Reference Materials

5.8.1 Participant's Handouts

1. Handouts on Integrated Bee Pastures

5.8.2. Training Reference Materials

1. Beekeeping in Kenya – Beginners guide (Ministry of Livestock Development 2011)
2. *Beekeeping: a beginner's guide* 1997. Baraka Agricultural College, Baraka, Kenya. (Carroll, T.)
3. *Bees for wealth and health: (Brochure)* Ministry of Agriculture & Rural Development, 2000 Nairobi, Kenya.

MODULE 6

STRATEGIC USE OF HONEYBEES FOR CROP POLLINATION

6.1 Introduction to the module

Insects and other animals pollinate about 87.5 % of world's flowering plants. Among the animal pollinators, bees are the major pollen vectors or pollinators. Recent reports have revealed that about three quarter of the crops benefit from pollination while one-third of the global food volume produced is as result of animal pollination. Pollinator abundance and diversity increases crop yields, quality and human nutrition. However, beekeepers in Africa including East Africa have not embraced the use of honeybee colonies for pollination; their focus has been mainly on production of hive products. However, in the effort to use honeybee colonies for pollination of crops, protection of bees from chemical exposure is an important factor. With the increasing dependence of pollinator dependent crops amidst declining bee populations, there is need to educate farmers on the use of managed bees for increased crop yields and quality.

6.2 Module learning outcomes

By the end of the module training the following training outcomes should be achieved:

- Strategic use of honeybee colonies for crop pollination identified.
- Impact of increased pollinator force on crop yields and quality quantified and explained.
- Protection of bees from chemical exposure mapped and discussed.
- Competition for pollinators with other crops and wild plants demonstrated and discussed.
- Carrying capacity of bee colonies per acre explained.

6.3 Module target group

This module targets service providers who include county extension staff and private service providers and lead farmers

6.4 Module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT) and farmer trainers. The trainees using this module should thoroughly familiarize themselves with the participant's handouts and training reference materials

6.5 Module duration

The module is estimated to take a duration of 4 hours 10 minutes

6.6 Module summary

Strategic use of honeybees for crop pollination			
Sessions	Training Methods	Training Materials	Time
6.6.1 Introduction and levelling of expectations	• Group discussion • Presentation	• PowerPoint Projector • Copy of Module Objectives • Felt pens, masking tape or sticker glue, notebooks and pens	20 minutes
6.6.2 Introduction to pollination ecology and crop pollination	• Presentation • Plenary discussion	• PowerPoint Projector, • Photos • Flip charts • Felt pens • Participants' Handouts	45 minutes
6.6.3 Assessment of pollination deficit	• Presentation • Class demonstration	• PowerPoint Projector, • Photos • Flip charts • Felt pens • Participants' Handouts	45 minutes
6.6.4 Honeybee pollination and environmental factors	• Presentation • Field demonstration	• PowerPoint Projector • Flip charts • Felt pens • Participants' handout	40 minutes
6.6.5 Handling of honeybee colonies and placement of colonies in the field	• Presentation • Plenary discussion	• PowerPoint Projector • Flip charts • Felt pens • Participants' handouts	40 minutes
6.6.6 Evaluating impact of introduced honeybee colonies on crop production.	• Presentation • Plenary discussion	• PowerPoint Projector • Flip charts • Felt pens • Participants' handouts	40 minutes
6.6.7 Module review	• Participants' Facilitator's summary	• PowerPoint Projector • Flip charts • Felt pens • Participants' Handout on module review	20 minutes
TOTAL			4 hours 10 minutes

6.7 Facilitator's Guidelines

Strategic use of honeybees for crop pollination	
6.7.1 Introduction and Levelling Expectations (20 minutes)	Session Guide
Introduction <i>(The facilitator welcomes trainees to the module and invites them to introduce themselves and state their expectation for the module).</i> <i>The facilitator then presents the module objectives</i> Module Objectives By the end of the module the trainee should be able to: • Identify strategic use of honeybee colonies for crop pollination. • Quantify and explain impact of increased pollinator force on crop yields and quality. • Map and discuss protection of bees from chemical exposure. • Demonstrate and discuss competition for pollinators with other crops and wild plants. • Explain carrying capacity of bee colonies per acre.	Summarize trainees' "expectations" and display them PowerPoint presentation and discussion Distribute Participant Handouts on Module Objectives.
6.7.2 Introduction to pollination ecology and crop pollination (45 minutes)	
Plenary Presentation (30 minutes) <i>(The facilitator describes the recommended methods of honey harvesting and required equipment)</i> • Bee pollination • Crops aided by bees in pollination (horticultural crops), • Quantity of food produced through bee pollination Discussion (15 minutes) Let the trainees recall what they learned and discuss any issues that may arise	PowerPoint presentation and discussion Distribute Participants' handout
6.7.3 Assessment of pollination deficit (45 minutes)	
Plenary Presentation (30 minutes) <i>(The facilitator guides the trainees in understanding the Floral sources of honey and flowering phenology).</i> • Level of pollination service • pollinator density • pollinator diversity in a landscape • Agronomic yield Discussion (15 Minutes) After the presentations allow trainees to raise any issues and discuss them	PowerPoint presentation and discussion Distribute Participants' handout

6.7.4 Honeybee pollination and environmental factors (40 minutes)	
Plenary Presentation (30 minutes) <i>(The facilitator leads the trainees in highlighting the importance of managing bee pastures).</i> • Environmental determinants in honeybee pollination • Climate change and weather variability • Reseeding and restoration of degraded landscapes Discussion (10 minutes) After the presentations allow trainees to raise any issues and discuss them	PowerPoint presentation and discussion Distribute Participants' handout
6.7.5 Handling of honeybee colonies and placement of colonies in the field (40 minutes)	
Plenary Presentation (25 minutes) <i>(The facilitator leads the trainees in aspects of handling bees and apiary siting).</i> • Basic Beekeeping operations • Hive location • Hive inspection • Requeening • Feeding bees in dearth periods Discussion (15 minutes) After the presentations, allow trainees to raise any issues and discuss them.	PowerPoint presentation and discussion Distribute Participants' handout
6.7.6 Evaluating impact of introduced honeybee colonies on crop production. (40 Minutes)	
Plenary Presentation (25 minutes) <i>(The facilitator leads the trainees in aspects of enhanced food production through bee pollination).</i> • Bees foraging on crops • honeybee colony quality on crop yield • pollinator benefit • Strong honeybee colony Discussion (15 minutes) After the presentations, allow trainees to raise any issues and discuss them.	PowerPoint presentation and discussion Distribute Participants' handout

6.7.7 Module review (20 minutes)	
<i>(The facilitator should let the trainees present their views on each of the sessions covered under this module. On flip charts list and summarize the key points they should emphasize when training farmers).</i> • Review together with the trainees the main points Strategic use of honeybees for crop pollination What new things did you learn from this Module? • What are some of the issues that you have become more aware of? • What questions do you still have?	Recap of the key take home points using any of the following participatory methods: • Q & A session • Discussions • Any other Distribute Participants' handout on module review

6.8 Training Notes and Reference Materials

6.8.1 Participants' Handouts

1. Handouts on strategic use of honeybees for crop pollination

6.8.2. Training Reference Materials

1. Beekeeping in Kenya – Beginners guide (Ministry of Livestock Development 2011)
2. *Beekeeping: a beginner's guide* 1997. Baraka Agricultural College, Baraka, Kenya. (Carroll, T.)
3. *Bees for wealth and health: (Brochure)* Ministry of Agriculture & Rural Development, 2000 Nairobi, Kenya.

MODULE 7

GENDER ROLES IN BEEKEEPING

7.1 Introduction to the Module

Traditionally men are involved in beekeeping activities which include; apiary cleaning, transportation of hives, harvesting, marketing of honey and offering security to the hives. On the other hand, the roles of women in beekeeping include; transportation of hives and provision of water. Exclusion of women from handling financial gains was attributed to cultural reasons such as the man being the household head. Therefore, he makes decisions concerning finances received from every household investment.

7.2 Module learning outcomes

By the end of the module training the following training outcomes should be achieved:

- Women involvement and gender roles in beekeeping explained and discussed
- Gender roles in beekeeping at the farm level mapped and discussed
- Household decision making on honey production activities identified
- Men and women participation in Beekeeping discussed
- Beekeeping as a business explained
- Gender awareness, lobbying and advocacy created.

7.3 Module target group

This module targets service providers who include county extension staff and private service providers and lead farmers.

7.4 Module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT) and farmer trainers. The training using this module should thoroughly familiarize themselves with the participants' handouts and training reference materials

7.5 Module duration

The module is estimated to take a duration of 4 hours 15 minutes

7.6 Module summary

Gender Roles in Beekeeping			
Sessions	Training Methods	Training Materials	Time
7.6.1 Introduction and levelling of expectations	• Presentation • Group discussion	• PowerPoint Projector • Copy of Module Objectives	20 minutes
7.6.2 An overview of women involvement and gender roles in beekeeping	• Presentation • Discussion	• PowerPoint Projector, Photos • Flip charts • Felt pens • Participants' Handouts	45 minutes
7.6.3 Gender roles in beekeeping at the farm level	• Presentation • Class demonstration	• PowerPoint Projector, Photos • Flip charts • Felt pens • Participants' Handouts	30 minutes
7.6.4 Household decision making on honey production activities	• Presentation • Field demonstration	• PowerPoint Projector • Flip charts • Felt pens • Participants' Handouts	30 minutes
7.6.5 Men and women participation in Beekeeping	• Presentation • Discussions	• PowerPoint Projector • Flip charts • Felt pens • Participants' handouts	30 minutes
7.6.6 Beekeeping as a business	• Presentation • Discussion	• PowerPoint Projector • Flip charts • Felt pens • Participants' handouts	50 minutes
7.6.7 Gender awareness lobbying and advocacy	• Presentation • Discussion	• Participants' Flip charts • Felt pens • Handout	30 minutes

7.6.8 Module review	• Participants' questions and comments • Facilitator's summary	• Flip charts • Felt pens • Participants' Handout for module review	20 minutes
TOTAL			4 hours 15 minutes

7.7 Facilitator's Guidelines

Gender Roles in Beekeeping	
7.7.1 Introduction and Levelling Expectations (20 minutes)	Session Guide
Introduction <i>(The facilitator welcomes trainees to the module to introduce themselves and state their expectations for the module).</i> <i>The facilitator then presents the module objectives</i> Module Objectives By the end of the module the trainee should be able to: • Explain and discuss women involvement and gender roles in beekeeping. • Map and discuss gender roles in beekeeping at the farm level. • Identify and explain decision making on honey production activities. • Discuss men and women participation in beekeeping. • Explain beekeeping as a business. • Create gender awareness, lobbying and advocacy.	Summarize trainees' "expectations" and display them Distribute Participants' Handouts on Module Objectives.
7.7.2 An overview of women involvement and gender roles in Beekeeping (45 minutes)	
Plenary Presentation (30 minutes) <i>(The facilitator should be able to guide the trainees in understanding the gender roles in beekeeping)</i> • Gender roles • opportunities for all genders in beekeeping, and the associated benefits Discussion (15 minutes) Let the trainees recall what they learned and discuss any issues that may arise	PowerPoint presentation and discussion Distribute Participants' handout

7.7.3 Gender roles in beekeeping at the farm level (30 minutes)	
Plenary Presentation (30 minutes) <i>(The facilitator guides the trainees in understanding the Floral sources of honey and flowering phenology).</i> • Gender roles in beekeeping • Assessing differences in gender roles and women empowerment • Household decision making on honey production activities • Empowerment of women, youth and the VMGs Discussion (10 Minutes) After the presentations allow trainees to raise any issues and discuss them	PowerPoint presentation and discussion Distribute Participants' handout
7.7.4 Household decision making on honey production activities (30 minutes)	
Plenary Presentation (20 minutes) <i>(The facilitator leads the trainees in highlighting the importance household involvement in beekeeping activities).</i> • Beekeeping as a livelihood activity • Support services to promote beekeeping Discussion (10 minutes) After the presentations, allow participants to raise any issues and discuss them.	PowerPoint presentation and discussion Distribute Participants' handout
7.7.5 Men and women participation in Beekeeping (30 minutes)	
Plenary Presentation (25 minutes) <i>(The facilitator leads the trainees in understanding the importance of participation of men and women in beekeeping).</i> • Beekeeping as rural livelihood strategy • Livelihood diversification • Cultural/Social factors hindrances Discussion (15 minutes) After the presentations, allow trainees to raise any issues and discuss them.	PowerPoint presentation and discussion Distribute Participants' handout

7.7.6 Beekeeping as a business (50 Minutes)	
Plenary Presentation (30 minutes) <i>(The facilitator leads the trainees in aspects the commercial benefits derived from beekeeping)</i> • From honey to money • What is the growth potential for beekeeping business • Beekeeping business plan Discussion (20 minutes) After the presentations, allow trainees to raise any issues and discuss them.	PowerPoint presentation and discussion Distribute Participants' handout
7.7.7. Gender awareness lobbying and advocacy (30 minutes)	
Plenary Presentation (20 minutes) <i>(The facilitator leads the trainees in aspects of gender mainstreaming in beekeeping).</i> • Gender awareness • Advocacy and lobbying for equal representation Discussion (10 minutes) After the presentations, allow trainees to raise any issues and discuss them.	PowerPoint presentation and discussion Distribute Participants' handout
7.7.7 Module review (20 minutes)	
<i>(The facilitator should let the trainees present their views on each of the sessions covered under this module. On flip charts, list and summarize the key points they should emphasize when training farmers)</i> Review together with the trainees the main points in Gender Roles in Beekeeping • What new things did you learn from this Module? • What are some of the issues that you have become more aware of? • What questions do you still have?	Recap of the key take home points using any of the following participatory methods: • Q & A session • Discussions • Any other Distribute participants' handout on module review

7.8 Training Notes and Reference Materials

7.8.1 Participant's Handouts

1. Handouts on Gender Roles in Beekeeping

7.8.2. Training Reference Materials

1. Beekeeping in Kenya – Beginners guide ((Ministry of Livestock Development 2011)
2. *Beekeeping: a beginner's guide* 1997. Baraka Agricultural College, Baraka, Kenya. (Carroll, T.)
3. *Bees for wealth and health: (Brochure)* Ministry of Agriculture & Rural Development, 2000 Nairobi, Kenya.

MODULE 8

BEEKEEPING COST BENEFIT ANALYSIS

While there are opportunities to improve people's livelihood through beekeeping, the way it is done currently indicates that significant benefits accruing from this activity are not reaching the poor beekeeping households and processors are not getting enough good quality products for their business. This is as a result of multiplicity of constraints such as decline of the bee population, environmental degradation, inadequate skills/best practices in the production of high yields of good honey, inadequate access to good quality inputs, poor entrepreneurship skills among producers, inadequate capital for processors and packers to invest in the sector, limited capacity of the producers to understand markets, lack of honey collecting centres, poor market linkages between the packers and producers and above all, absence of effective grassroots/village-based producer groups capable of producing sufficient honey for the market and absence of a strong private sector.

8.2 Module learning outcomes

By the end of the module training the following training outcomes should be achieved:

- Commodity background identified and explained
- Production regions and volumes discussed and explained
- Honey volumes and price patterns discussed
- Project modelling - analytical framework demonstrated and discussed
- Expenditures (input and operating costs) and incomes demonstrated and explained
- Economic analysis demonstrated and discussed.

8.3 Module target group

This module targets service providers who include county extension staff and private service providers and lead farmers.

8.4 Module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT) and farmer trainers. The trainers using this module should thoroughly familiarize themselves with the participants' handouts and training reference materials.

8.5 Module duration

The module is estimated to take a duration of 3 hours 10 minutes

8.6 Module summary

Beekeeping Cost Benefit Analysis			
Sessions	Training Methods	Training Materials	Time
8.6.1 Introduction and levelling of expectations	• Presentation • Group discussion	• Copy of module Objectives • PowerPoint Projector	20 minutes
8.6.2 Commodity Background	• Presentation • Group discussion	• PowerPoint Projector, Photos • Flip charts • Felt pens • Participants' Handouts	40 minutes
8.6.3 Production Regions and Volumes	• Presentation • Class demonstration	• PowerPoint Projector, Photos • Flip charts • Felt pens • Participants' Handouts	40 minutes
8.6.4 Honey Volumes and Price Patterns	• Presentation • Field demonstration	• PowerPoint Projector • Flip charts • Felt pens • Participants' Handouts	30 minutes
7.6.5 Project modelling - Analytical Framework	• Presentation • Discussions	• PowerPoint Projector • Flip charts • Felt pens • Participants' handouts	20 minutes
8.6.6 Expenditures (Input and Operating Costs) and Incomes	• Presentation	• PowerPoint Projector • Flip charts • Felt pens • Participants' Handouts	20 minutes

8.6.7 Module review	• Participants' Facilitator's summary	• Flip charts • Felt pens • Participants' Handout on module review	20 minutes
TOTAL			3 hours 10 minutes

8.7 Facilitator's Guide

Beekeeping Cost Benefit Analysis	
8.7.1 Introduction and Levelling Expectations (20 minutes)	Session Guide
Introduction <i>(The facilitator welcomes trainees to the module and invite them to introduce themselves and state their expectation for the module).</i> <i>The facilitator then presents the module objectives</i> Module Objectives By the end of the module the trainee should be able to: • Identify and explain commodity background • Discuss and explain production regions and volumes • Discuss honey volumes and price patterns • Demonstrate and discuss project modelling - analytical framework • Demonstrate and explain expenditures (input and operating costs) and incomes • Demonstrated and discuss economic analysis	Summarize Participants' "Expectations" and display them Distribute Participant Handout on module objectives.
8.7.2 Commodity Background (40 minutes)	
Plenary Presentation (30 minutes) <i>(The facilitator guides the trainees in understanding the aspects of honey and its by-products)</i> • nature of honey, • hive products, • value addition Discussion (10 minutes) Let the trainees recall what they learned and discuss any issue that may arise	PowerPoint presentation and discussion Distribute participants' handout

8.7.3 Production Regions and Volumes (40 minutes)	
Plenary Presentation (30 minutes) <i>(The facilitator guides the trainees in understanding the quantity of honey and hive produced in different regions).</i> • Honey production • Hive products; Wax, propolis, bee venom and royal jelly Discussion (10 Minutes) After the presentations allow trainees to raise any issues and discuss them	PowerPoint presentation and discussion Distribute Participants' handout
8.7.4 Honey Volumes and Price Patterns (30 minutes)	
Plenary Presentation (20 minutes) <i>(The facilitator leads the trainees in understanding the honey volumes produced in different regions).</i> • Volumes of honey produced • Honey and other hive products Niche markets • Selling value of different hive products Discussion (10 minutes) After the presentations, allow trainees to raise any issues and discuss them.	PowerPoint presentation and discussion Distribute Participants' handout
8.7.5 Project modelling - Analytical Framework (20 minutes)	
Plenary Presentation (30 minutes) <i>(The facilitator should be able to lead the trainees in aspects the nature of honey business).</i> • Beekeeping as a business • Business model establishment • Customer profile • Strong marketing skills Discussion (10 minutes) After the presentations, allow trainees to raise any issues and discuss them.	PowerPoint presentation and discussion Distribute Participant's handout

8.7.6. Expenditures (Input and Operating Costs) and Incomes (20 minutes)	
Plenary Presentation (30 minutes) <i>(The facilitator should be able to lead the trainees in aspects of understanding input and operating costs).</i> • Input costs • Operating cost/cost of production • Profit and Loss in a business enterprise • Doing beekeeping as a business Discussion (10 minutes) After the presentations allow trainees to raise any issues and discuss them	Distribute participant's handout on cost benefit analysis
8.7.6 Module review (20 minutes)	
Plenary Presentation and Discussion (20 minutes) <i>(The facilitator should let the trainees present their views on each of the sessions covered under this module. On flip charts, list summarize the key points they should emphasize when training farmers).</i> • Review together with the trainees the main points in Beekeeping Cost Benefit Analysis • What new things did you learn from this Module? • What are some of the issues that you have become more aware of? • What questions do you still have?	Recap of the key take home points using any of the following participatory methods: • Q & A session • Discussions • Any other Distribute Participants' handout on module review

8.8 Training Notes and Reference Materials

8.8.1 Participant's Handouts

1. Handouts on Beekeeping Cost Benefit Analysis

8.8.2. Training Reference Materials

Beekeeping in Kenya – Beginners guide (Ministry of Livestock Development 2011)

1. *Beekeeping: a beginner's guide* 1997. Baraka Agricultural College, Baraka, Kenya. (Carroll, T.)
2. *Bees for wealth and health: (Brochure)* Ministry of Agriculture & Rural Development, 2000 Nairobi, Kenya.

ANNEX 1: TRAINING PROGRAM

The training program presented here assumes that the trainees report on Sunday evening as the first day and leave 8 days later on Sunday morning.

Time	Day 0 (Sunday)	Duration	Remarks / Facilitator
Late Evening	- Arrival of participants and registration – Host - Setting up and prepare training venue and materials	2 Hours	The training venue and materials are ready for use
<i>Close of Day 0</i>			
Time	Day 1 (Monday)	Duration	Remarks / Facilitator
8.30-10.30 am	Welcome by host and Prayers	10 minutes	The trainees relax and climate set for the ten-day training
	Self-introductions –(CTT)	30 minutes	
	Introduction to training on Animal Health Management (KALRO)	15 minutes	
	Official opening Ceremony (CDL/CDVS)	15 minutes	
	Introduction to the training program (CTT)	20 minutes	
	Formation of groups and setting norms (CTT)	30 minutes	
10.30-11.00 am	Tea Break (Group Photo)	30 minutes	Health Break
11.00 am – 1.00 pm	Module 1. Up scaling of beehive Technologies		
	- Climate setting and Introduction to the module	20 minutes	
	- Terminologies used in Beekeeping	15 minutes	
	An overview of beekeeping sector in Kenya	40 minutes	
	- Importance of beekeeping	45 minutes	

1.00-2.00 pm	Lunch Break	1 hour	
2.00-4.00 pm	Module 1. Up scaling of beehive Technologies (Cont') · Honeybee species and races of bees (60 minutes) · Common types of beehives in Kenya (60 minutes)	60 minutes 60 minutes	
4.00-4.30 pm	Coffee Break	30 minutes	
4.30 – 5.10 pm	Module 1. Beekeeping and Beehive technologies (cont.) · Advantages and disadvantages of different beehives · Conclusions and review of day's topics	30 minutes 10 minutes	
Close of Day 1			

	Day 2 (Tuesday)	Duration	Remarks / Facilitator
8.30-10.30 am	Recap on Module 1. and levelling expectations Module 1. Beekeeping and Beehive technologies (cont') · Hive Production levels Module 2. Bee Forage and Ecosystem Conservation · Levelling expectations, module objectives · Types of bee forages in different AEZs · Bee forage conservation	10 minutes 40 minutes 10 minutes 30 minutes 30 minutes	
10.30-11.00 am	Tea Break	30 minutes	Health Break

11.00 am-1.00 pm	Module 2. Bee foraging and ecosystem conservation (cont.) · Bee pollination and its significance · Conclusions and review of module 2 Module 3. Housing of Bee Colonies and Types of Housing Technologies · Importance of housing of apiaries · Siting of apiaries	30 minutes 10 minutes 30 minutes 30 minutes 30 minutes	
Time	Day 3 (Wednesday)	Duration	Remarks / Facilitator
1.00-2.00 pm	Lunch Break	1 hour	
2.00-4.00 pm	Module 3. Housing of bee colonies and types of housing technologies (cont') · Regular beehive inspection and hygiene · Disease and pest management · Supplementary Feeding	40 minutes 60 minutes 20 minutes	
4.00-4.30 pm	Coffee Break	1 hour	
4.30-5.25 pm	Module 4. Harvesting, post harvesting handling and processing · Levelling expectations, module objectives · Honey harvesting and equipment · Conclusions and review of the day's modules	15 minutes 30 minutes 10 minutes	
Close of Day 3			

Time	Day 4 (Thursday)	Duration	Remarks / Facilitator
8.30am- 10.30am	Module 4. Harvesting, post harvesting handling and processing (cont.) - Recap - Harvesting procedure - Hive products	15 minutes 55 minutes 50 minutes	
10.30 -11.00 am	Tea Break	30 minutes	Health Break
11.00 am- 1.00 pm	Module 4. Harvesting, post harvesting handling and processing (cont') - Honey processing and packaging - Hive products value addition - Conclusions and review of module 4	50 minutes 50 minutes 20 minute	
1.00-2.00 pm	Lunch Break	1 hour	
2.00-4.00 pm	Module 5. Integrated Bee Pastures - Levelling expectations, module objectives - Introduction to bee diversity: managed and solitary bees. - Introduction to major bee floral resources and flowering phenology - Conservation of natural bee pastures and restoration of degraded habitats	15 Minutes 45 minutes 30 minutes 30 minutes	
4.00-4.30 pm	Coffee Break	30 minutes	

4.30 – 5.25 pm	- Management of bee pastures (Induced disturbance, reseeding, planting seedlings etc) - Conclusions and review of module 4	40 minutes	
Close of Day 4			
Time	Day 5 (Friday)	Duration	Remarks / Facilitator
8.30-10.30 am	Module 5. Integrated Bee Pastures (cont') - Threats to bee pastures and mitigation strategies - Module 5 review - Module 6. Strategic Use of Honeybees for Crop Pollination - Levelling expectations, module objectives - Assessment of pollination deficit	40 minutes 20 minutes 15 Minutes 45 minutes	
10.30 am-11.00 am	Tea Break	30 minutes	Health Break
11.00 am-1.00 pm	- Honeybee pollination and environmental factors - Handling of honeybee colonies and placement of colonies in the field - Evaluating impact of introduced honeybee colonies on crop production	40 minutes 40 minutes 40 Minutes	
1.00 -2.00 pm	Lunch Break	1 hour	

2.00-4.00 pm	Module 7. Gender Roles in Beekeeping · Levelling expectations, module objectives · An overview of women involvement and gender roles in Beekeeping · Gender roles in beekeeping at the farm level · Household decision making on honey production activities	15 Minutes 45 minutes 30 minutes 30 minutes	
4.00-4.30 pm	Coffee Break	30 minutes	
4.30 - 5.10 pm	· Men and women participation in Beekeeping · Conclusions and review of module 7	30 minutes 10 minutes	
Close of Day 5			
Time	Day 6 (Saturday)	Duration	Remarks / Facilitator
8.30-10.30 am	Module 7. Gender Roles in Beekeeping (cont) • Recap • Doing beekeeping as a business • Gender awareness lobbying and advocacy • Module review	20 minutes 50 minutes 30 minutes 20 minutes	
10.30-11.00 am	Tea Break	30 minutes	Health Break

11.00 am-1.00 pm	Module 8. Beekeeping Cost Benefit Analysis Levelling expectations, module objectives Commodity Background Production Regions and Volumes Honey Volumes and Price Patterns	10 minutes 40 minutes 40 Minutes 30 minutes	
1.00 -2.00 pm	Lunch Break	1 hour	
2.00-3.30 pm	Module 8. Beekeeping Cost Benefit Analysis (cont) · Project modelling - Analytical Framework · Expenditures (Input and Operating Costs) and Incomes · Conclusions and review of module 7	20 minutes 20 minutes 20 minutes	
3.30 -3.50 pm	Coffee Break	20 minutes	
3.50 – 5.00 pm	Course evaluation (individual and group evaluation) Official closing remarks Logistics	40 minutes 20 minutes	CTT CDA CTT
Close of Day 6			
Departure	Day 7 (Sunday)		

ANNEX 2: GENERAL REFERENCE MATERIALS

Category / Modules	Publication title	Reference types	No Pages	Farmer Category A= New entrant/ Beekeeper B= Established Beekeeper Farmer
1. Terminologies used in Beekeeping	Beekeeping Manual	Manual		A/B
				A/B
2. Beekeeping and hive products	Guardian bees in Kenya	Journal	8	A/B
3. Apiary management	Beekeeping for Beginners'	Guide		A/B
4. Pest and disease management				
Marketing of honey and hive products				



Kenya Climate Smart
Agriculture Project

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